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Small extracellular vesicles derived from mesenchymal stromal cells loaded with β -nicotinamide mononucleotide activate NAD^+ /SIRT3 signaling pathway-mediated mitochondrial autophagy to delay skin aging

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Abstract

Background Recently, the beneficial effects of human umbilical cord mesenchymal stromal cell (hucMSC)-derived small extracellular vesicles (sEVs) in mitigating skin aging through multiple mechanisms have been widely reported. β -Nicotinamide mononucleotide (NMN) is an iconic anti-aging drug that increases NAD^+ levels in the body to slow down, ameliorate, and prevent various phenotypes associated with aging, but its high water solubility, low permeability, and instability limit its clinical application. Based on this, we applied electroporation to construct NMN-loaded hucMSC-sEVs (NMN-sEVs) to improve their stability and efficacy and to enhance their potential for translational application in medical aesthetics and anti-aging.

Methods D-galactose was applied to construct a mouse skin aging model, based on which comparative analyses of topical and nano-microneedle administration were performed to determine the optimal delivery method of sEVs in vivo experiments. After constructing NMN-sEVs by electroporation, high-performance liquid chromatography was applied to detect the loading efficiency, and the effects of NMN-sEVs on delaying skin aging were assessed by histological analysis. In addition, the defense effects of NMN-sEVs against cellular senescence were verified by reactive oxygen species assay, β -galactosidase staining, qRT-PCR, Western blot, and cellular immunofluorescence. Finally, the roles of NMN-sEVs in remodeling mitochondrial function and delaying cellular senescence through mitochondrial autophagy were assessed by mitochondrial mass, function, and autophagy level assays.

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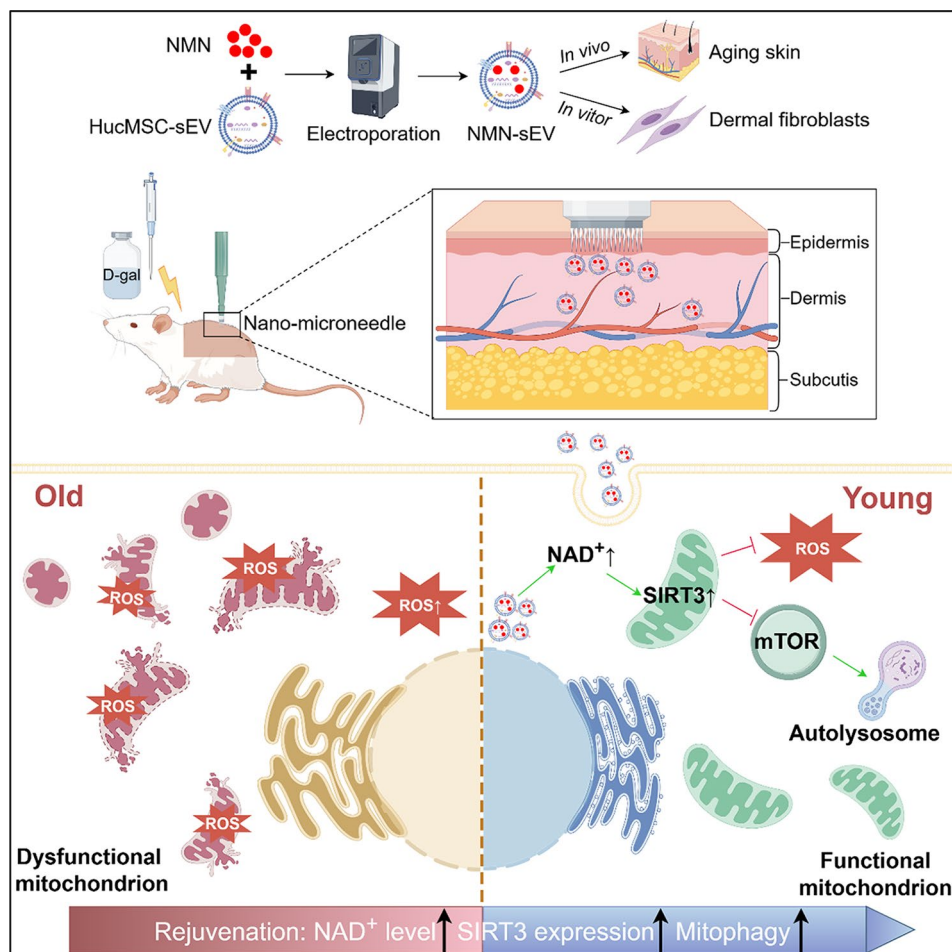


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Results Our data suggested that NMN-sEVs could improve skin aging in mice, delay cellular senescence, and restore cellular mitochondrial dysfunction. Notably, NMN-sEVs treatment increased intracellular NAD^+ levels and SIRT3 expression, as well as rescued the inhibition of senescence-induced mitochondrial autophagy, suggesting a role for NMN-sEVs in the remodeling of mitochondrial function through mitochondrial autophagy. Additionally, the use of the SIRT3 inhibitor 3-TYP suppressed the positive effects of NMN-sEVs on cellular senescence, mitochondrial function, and mitochondrial autophagy while restoring senescence-associated characteristics.

Conclusion Overall, our findings revealed a mechanism by which NMN-sEVs attenuated mitochondrial dysfunction and rescued cellular senescence by promoting NAD^+ /SIRT3 pathway-mediated mitophagy and might provide a promising strategy for anti-aging pharmaceuticals.

Graphical Abstract



Keywords HucMSC-sEVs, NMN, Skin aging, SIRT3, Mitochondrial autophagy

Introduction

As we age, all of the body's organs undergo deterioration, with skin aging being particularly evident [1]. As the body's primary superficial organ, the skin maintains continuous interaction with ambient surroundings while performing essential barrier functions against dehydration, ultraviolet (UV) radiation, and pathogens [2, 3]. Skin aging is influenced by both internal and external factors,

leading to a deterioration in the skin's structural integrity and a reduction in its physiological functions [4]. Endogenous skin aging occurs due to gradual changes that lead to a buildup of cellular damage, resulting in thinner skin and the appearance of fine lines [5]. Exogenous skin aging is linked to lifestyle and environmental exposures that contribute to aging, with photo-aging resulting from UV radiation being the most prevalent factor [6].

Glycosylation is another key factor in skin aging that has received attention in recent years in addition to UV radiation [7]. D-galactose (D-gal), as a powerful glycosylation agent, produces advanced glycation end products (AGEs) in the organism and accumulates in the skin, which disrupts cellular function by inducing oxidative stress and ultimately accelerates skin aging [8]. Therefore, D-gal has been extensively utilized in aging models for anti-aging research worldwide [9]. Notably, the skin has a cutaneous neuroendocrine system that can be activated by environmental or endogenous stressors and produce neuroendocrine mediators to maintain communication between the skin and the brain. A study showed that after exposure of mouse skin to UV radiation, the cutaneous hypothalamic-pituitary-adrenal axis increased post-exposure cortisol levels, induced a stress response in the hippocampus, impaired neurogenesis, and decreased synaptic protein expression, suggesting that the skin affected brain function [10]. In addition, as senescence occurs, senescent skin can transmit senescence from the skin to other tissues and organs through the secretion of senescence-associated secretory phenotypes (SASPs), thereby affecting whole-body senescence [11]. SASP genes are upregulated during senescence, and senescent cells drive whole-body senescence through the SASP network regulated by NF- κ B or C/EBP β [12]. Others, such as miRNAs or the rapamycin receptor (mTOR) pathway, also play a role in SASP regulation [13]. A recent study showed that the presence of senescent cells in the dermis of young mice resulted in increased proximal and distal host tissue senescence, along with increased frailty, impaired musculoskeletal function, and a significant decline in cognitive function, as well as an increase in the expression of senescence-associated markers in hippocampal brain regions [14]. Therefore, targeting skin senescence through topical administration of anti-aging drugs may contribute to the development of new anti-aging strategies to delay the onset of systemic aging and age-related diseases. Currently, emerging anti-aging strategies target the elimination of senescent cells and the modulation of SASPs [15–18]. Senolytics, drugs that selectively eliminate senescent cells, have been applied topically to the skin, showing potential anti-aging effects. In addition, both metformin and rapamycin have been applied topically to human skin [19, 20]. Rapamycin significantly reduced P16 levels, increased collagen VII, and showed an overall improvement in visible skin structure [19]. However, more studies are still needed to better elucidate the efficacy of these anti-aging agents on skin aging as well as systemic aging and the role of skin aging in systemic aging.

Research has demonstrated that the buildup of senescent cells in the epidermis, dermis, and subcutaneous tissue leads to skin aging [21]. These senescent cells stop

proliferating but can remain metabolically active and induce peripheral cellular senescence through the secretion of SASPs, such as inflammatory factors, chemokines, extracellular vesicles (EVs), matrix metalloproteinases (MMPs), and growth factors, and lead to persistent chronic inflammation (inflammaging), resulting in tissue dysfunction as well as the emergence of a senescent phenotype [22–25]. Senescent cells are also featured by telomere shortening, metabolic changes, mitochondrial dysfunction, and reduced autophagy [22, 26, 27]. Mitochondrial dysfunction represents a significant characteristic of cellular senescence. Dysfunctional mitochondria result in impaired mitochondrial integrity and biogenesis, diminished mitochondrial membrane potential, elevated levels of reactive oxygen species (ROS), and ultimately accelerated cellular senescence or individual aging [28–32]. Similarly, mitochondrial dysfunction disrupts redox homeostasis, leading to cytoplasmic NADH accumulation, and decreased NAD⁺/NADH ratios result in ATP exhaustion and cell cycle arrest [31, 33]. In addition, NAD⁺ levels and autophagic activity decline with senescence, leading to impaired communication between mitochondria and the nucleus as well as the accumulation of damaged organelles, including mitochondria, whereas mitochondrial elimination induced by mitochondrial autophagy ameliorates cellular senescence and inhibits the secretion of SASP factors [34–37]. It has been shown that autophagy and mitochondrial quality control can be enhanced by NAD⁺ supplementation, thereby promoting tissue rejuvenation and organismal longevity. Drivers of this process are NAD⁺-depleting enzymes, such as sirtuin family deacetylases (SIRT3) [38]. Among them, SIRT3, which is present inside mitochondria, has been documented to promote mitochondrial autophagy via the mitochondrial SIRT3-PGAM5-FUNDC1 pathway [39]. Therefore, supplementing NAD⁺ precursors like β -nicotinamide mononucleotide (NMN) to increase NAD⁺ levels and thus promote mitochondrial autophagy is of broad guidance in the quest to delay cellular senescence.

Stem cell therapies, represented by mesenchymal stromal cells (MSCs), have received significant interest in tissue regenerative medicine research in recent years. MSCs can promote the repair of damage in various tissues and organs, especially showing significant efficacy in the treatment of some clinically complex and refractory diseases like diabetes and spinal cord injuries [40, 41]. As an important component of MSC paracrine secretion, EVs retain the inherent regenerative capabilities of MSCs. Furthermore, they exhibit enhanced stability and reduced immunogenicity, which makes them extensively applied in tissue regeneration following diverse forms of injuries. EVs are nano-sized vesicles wrapped in a lipid bilayer, capable of being secreted by nearly all cell types,

and are abundant in various bioactive substances. Based on particle size, the classifications include small EVs (sEVs, <200 nm) and large EVs (lEVs, >200 nm) [42]. Our early research demonstrated that human umbilical cord MSC (hucMSC)-derived sEVs (hucMSC-sEVs) containing 14-3-3 ζ protein have the potential to accelerate ROS clearance, inhibit DNA damage, and suppress apoptosis by upregulating SIRT1 in keratinocytes, thereby attenuating UV radiation- and H₂O₂-induced oxidative stress damage [43]. In addition, hucMSC-sEVs could positively regulate pregnancy zone protein to inhibit the expression of MMP1, the main enzyme in the MMP family that degrades type I collagen, thereby preventing UV-induced skin photoaging [44]. However, the roles of natural sEVs in delaying skin aging are limited, and engineering modifications such as anti-aging drugs can enhance the efficacy of sEVs. NMN, as a precursor to NAD⁺, enhances autophagy and normalizes mitochondrial function by increasing NAD⁺ levels, thereby delaying, improving, and preventing various aging-related phenotypes, as well as age-induced metabolic disorders and age-related diseases [45, 46]. However, NMN has high water solubility, low permeability, and instability. Considering the potential of NMN in delaying skin aging and the need to enhance the therapeutic effects of sEVs, modifying sEVs with NMN is an effective strategy to couple the advantages of both.

In this study, we developed an NMN nano-drug delivery system and identified the beneficial effects of NMN-loaded hucMSC-sEVs (NMN-sEVs) in delaying skin aging. We have confirmed that NMN-sEVs could improve skin aging by enhancing NAD⁺ levels to promote mitochondrial autophagy. Furthermore, as a NAD⁺-dependent deacetylase, SIRT3 plays a crucial role in the regulation of mitochondrial function and the modulation of mitochondrial autophagy. In summary, our research examined the regulatory influences and potential mechanisms by which NMN-sEVs affect mitochondria and aging, revealing a therapeutic pathway that links NMN, sEVs, autophagy, mitochondrial function, and skin aging.

Materials and methods

Cell isolation and culture

With the approval of the Ethics Committee of the Fourth Affiliated Hospital of Jiangsu University, primary hucMSCs were obtained according to the previously established tissue explants adherent method [47]. In brief, the umbilical cord's vascular tissue was removed first. Then the remaining tissue was diced into 2 mm³ fragments and arranged uniformly in a culture dish (Corning). The petri dishes were inverted and placed in a cell culture incubator (Thermo Fisher) for approximately 60 min. After the tissue blocks adhered, minimum essential medium α (α -MEM, Gibco) containing 15% fetal bovine serum (FBS, Gibco) and 1% penicillin-streptomycin

(Bioind) was added for primary culture. The spent culture medium was changed every 3 days for approximately 10 days. Carefully removed the umbilical cord tissue when a large number of cells appeared at the edges of the umbilical cord tissue and grew densely in a swirling pattern. Cells were collected by trypsin digestion and transferred to α -MEM medium containing 10% FBS for further expansion, and the resulting cells were 1st generation hucMSCs.

EVs were removed from FBS by centrifugation at 4 °C, 100,000 g for 16 h (EVs-depleted FBS). The hucMSCs were collected by trypsin digestion and transferred to α -MEM medium containing 10% FBS for cell culture. When the fusion of hucMSCs reached 50–70%, the cells were washed three times with phosphate-buffered saline (PBS, Meilunbio) and replaced with α -MEM medium containing 10% EVs-depleted FBS, and then the supernatant was collected after 48 h of incubation. The above steps were repeated, and the supernatants from the 3rd to the 7th generation of cells were collected for the extraction of hucMSC-sEVs.

Dermal fibroblasts (DFs) were primarily isolated from the dorsal skin of 1–2 day-old SD rats. The dorsal skin of SD rats was cut and scraped to remove blood vessels and adipose tissue. The dermis and epidermis were gently separated after the addition of 0.25% Dispase enzyme (Sigma) for overnight digestion at 4 °C. The dermis was cut into 1–2 mm³ pieces and evenly affixed to the bottom of the petri dish. Subsequently, they were cultured under conditions of 37 °C and 5% CO₂ for approximately 7 days. The spent culture medium was replaced every 3 days during this period using α -MEM containing 10% FBS and 1% penicillin-streptomycin to maintain growth. Generation 3 to 5 cells were used for subsequent cell experiments. DFs were identified by Western blotting and cellular immunofluorescence using α -SMA, vimentin, and CK10.

Separation, identification, and tracking of hucMSC-sEVs

HucMSC-sEVs were extracted using differential ultracentrifugation [48]. The collected cell supernatants were centrifuged at 300 g for 10 min, 2000 g for 10 min, and 10,000 g for 30 min at 4 °C to remove dead cells, cell debris, and organelles, respectively. The supernatants were transferred to 100 kDa MWCO ultrafiltration centrifuge tubes (Millipore) and centrifuged at 2000 g for 30 min to concentrate the sample volume. The concentrates were centrifuged at 100,000 g for 120 min at 4 °C, and the white precipitates obtained were hucMSC-sEVs. The supernatants were discarded, and the precipitates were resuspended by adding PBS and then centrifuged at 100,000 g for 120 min at 4 °C. The supernatants were discarded, and the precipitates were dissolved by adding an appropriate amount of PBS overnight. The completely solubilized hucMSC-sEVs were sterilized by

a 0.22 μm filter (Millipore) and used for subsequent experiments. The size and morphology of the sEVs were observed using transmission electron microscopy (TEM, FEI Tecnai 12, Philips). The size distribution, concentration, and zeta potential of the sEVs were measured using a nanoparticle tracking analyzer (NTA, Particle Metrix). Western blotting was used to detect surface markers of hucMSC-sEVs.

CM-DiL membrane dye (Invitrogen) was co-cultured with hucMSC-sEVs in the dark for 1 h to obtain DiL-labeled sEVs. Unbound dye was then removed using 100 kDa MWCO ultrafiltration centrifuge tubes. After that, DiL-labeled sEVs were administered to the dorsal skin of the mice using nano-microneedles (Suzhou Natong Biological Nanotechnology Corporation Ltd.). After preparing the skin tissue into frozen sections, observed the fluorescence distribution using a fluorescence microscope. In vitro, DiL-labeled sEVs were co-cultured with DFs for 24 h. The uptakes of DiL-labeled sEVs by DFs were subsequently examined using fluorescence microscopies.

Skin barrier recovery experiment

The degree of action of nano-microneedles on the stratum corneum was evaluated by observing the closing speed of the pores of the stratum corneum. Mice were anesthetized (avertin, 1.25%, 30 $\mu\text{l/g}$, Meilunbio), and the hair on the back was removed and recovered for 24 h. 100 μL of 0.4% trypan blue dye (Pricella) was first evenly dripped onto the dehairing back skin of the mice, and then the nano-microneedles were applied to the skin of the mice for about 20 min. The excess of the trypan blue dye was blotted with a dry cotton ball, and then the residual dye was wiped off with a cotton ball soaked in 75% alcohol. The diffusion of trypan blue dye in the skin was observed at 0 min, 5 min, 10 min, 30 min, 1 h, 2 h, 4 h, 6 h, 12 h, and 24 h after application of the nano-microneedles, and the effect of the nano-microneedles on the stratum corneum was judged according to the time of fading of the skin staining.

Preparation of NMN-loaded hucMSC-sEVs

Using electroporation technology to prepare NMN-sEVs. In brief, a certain amount of hucMSC-sEVs were mixed with NMN (>98.0% HPLC, TCI), and the final concentration of hucMSC-sEVs was adjusted to 1×10^{10} particles/mL using PBS, while the final concentration of NMN was adjusted to 30 mM. After mixing thoroughly, added the mixture solution slowly to the electrode cup. The following electroporation parameters were set on the CUY21EDIT II Super Multi-Mode Gene Electroporator (BEX): decay mode, pore voltage 110 V, duration 3 ms, rest 10 ms; driving voltage 25 V, duration 50 ms, rest 50 ms; driving cycles 10, capacitance 940 μF . After adjusting

the solution resistance to a suitable degree, the operation was performed. Upon completion of electroporation, the mixture was incubated under conditions of 37 $^{\circ}\text{C}$ for 1 h. PBS was added to dilute, and a 100 kDa MWCO ultrafiltration centrifuge tube was used to concentrate and wash away the free NMN.

The encapsulation efficiency of NMN was determined using ultrafiltration centrifugation [49]. The mixture after electroporation was placed in a 100 kDa MWCO ultrafiltration centrifuge tube and centrifuged at 2000 g for 30 min, with NMN-sEVs retained in the ultrafiltration centrifuge tube. NMN levels in the filtrate were assessed using HPLC to determine encapsulation rate and drug loading.

D-gal-induced skin senescence model and treatment

Twenty-four female ICR mice (6 weeks old, weighing 24 ± 2 g) were purchased and housed at the Laboratory Animal Center of Jiangsu University. They were kept in cages at a temperature of 22 ± 2 $^{\circ}\text{C}$ and given adequate diet and water (12 h light/dark cycle). All animal experiments complied with standard guidelines for the care and use of laboratory animals and were approved by the Institutional Animal Care and Use Committee of Jiangsu University. One week prior to the formal experiment, the aforementioned ICR mice were fed adaptively and then randomly divided into four groups of six mice each: Control (CTR), D-gal (D-gal), D-gal + topical + sEVs (sEV + TOP), and D-gal + nano-microneedles + sEVs (sEV + Nano-microneedle). A D-gal-induced mouse skin aging model was established as described previously [50]. In brief, D-gal (MedChemExpress) was dissolved in 70% ethanol (10 mg/mL) and 0.5 mL was topically applied to the shaved back skin of the mice (4×2 cm^2) twice daily for 6 weeks. The shaved back skin (4×2 cm^2) of control mice was coated with 0.5 mL of 70% ethanol twice daily for 6 weeks to exclude the influence of 70% ethanol. On the first day of each week, after anesthetizing mice by intraperitoneal injection of 1.25% avertin, hucMSC-sEVs (5×10^9 particles/mouse/week) were administered via topical or nano-microneedles once a week for 6 weeks. At the end of the experiment, mice were euthanized under anesthesia via cervical dislocation, and skin tissues were collected for further examination. The experimenter was not the designer of the experiment to exclude the influence of subjective factors on the experimental results.

After successfully constructing a D-gal-induced mouse skin aging model and confirming the use of nano-microneedles as a transdermal drug delivery method, thirty-six 6-week-old female ICR mice were once again purchased and housed at the Laboratory Animal Center of Jiangsu University. The mice were randomly divided into six groups of six mice each: Control (CTR), D-gal (D-gal), D-gal + nano-microneedles + PBS

(PBS), D-gal + nano-microneedles + hucMSC-sEVs (sEV), D-gal + nano-microneedles + NMN (NMN), and D-gal + nano-microneedles + NMN-sEVs (NMN-sEV). A D-gal-induced mouse skin aging model was established as described previously. On the first day of each week, PBS, hucMSC-sEVs (5×10^9 particles/mouse/week), NMN (2.2 mg/mouse/week), and NMN-sEVs (5×10^9 particles/mouse/week) were administered locally via nano-microneedles once a week for 6 weeks. Since the D-gal-induced skin aging model mice had little effect on the health status of each mouse, all mice were included in the evaluation. At the end of the experiment, the Fitzpatrick Wrinkle Scale was used to assess the severity of skin aging through descriptive language. The work has been reported in line with the ARRIVE guidelines 2.0.

In vitro, the experiments were divided into a normal control group (CTR), a D-gal model group (D-gal), a D-gal+sEV pretreatment group (sEV), a D-gal+NMN pretreatment group (NMN), and a D-gal+NMN-sEV pretreatment group (NMN-sEV). After digestion and resuspension of DFs, 6-well cell culture plates were inoculated with 4×10^4 cells per well. After the cells were attached to the wall, the sEV, NMN, and NMN-sEV groups were pretreated with 1×10^9 particles hucMSC-sEVs, 0.44 mg NMN, and 1×10^9 particles NMN-sEVs, respectively. After 24 h of pretreatment, the D-gal, sEV, NMN, and NMN-sEV groups were stimulated for another 24 h by adding 1 mL of 10 mg/mL D-gal solution per well, respectively.

Immunohistochemical (IHC) staining

In brief, skin tissue sections were dewaxed, antigenically repaired, blocked, and incubated overnight with diluted primary antibody. Diluted secondary antibodies were added and incubated for 30 min. The SABC reagent (Boster) was then added and incubated in a humid chamber for 30 min. DAB chromogenic solution was added, and when a brown or brownish color appeared under the microscope, the sections were immediately placed in double-distilled water and rinsed. Hematoxylin staining was applied for 1 min, followed by rinsing under running water for 15 min. After dehydration, a drop of neutral resin was added, and a cover slip was placed on top. The primary antibodies used were as follows: P53 (1:100, Zenbio), PCNA (1:100, CST), 53BP1 (1:100, Zenbio), and SIRT3 (1:100, Wanleibio).

Cellular immunofluorescence

The processed cells were fixed at room temperature with 4% paraformaldehyde, followed by membrane permeabilization at room temperature, and then blocked with 5% BSA for 2 h. The primary antibodies were incubated overnight under 4 °C conditions. The secondary antibodies were incubated under 37 °C conditions for 30 min.

Hoechst33342 dye (1:200, Sigma Aldrich) was used to stain cell nuclei. After applying an anti-fluorescence quencher, cell images were obtained under a microscope. The principal antibodies employed in this research were ki67 (1:100, Proteintech), P16 (1:100, Proteintech), and LaminB1 (1:100, Abcam).

Western blotting

Cells were lysed to extract total cellular proteins. After separating equal amounts of protein samples by electrophoresis, they were transferred to a PVDF membrane (Millipore). It was then closed with 5% skimmed milk for 1 h. The membrane was then incubated with the primary antibodies overnight under 4 °C conditions. The membrane was then incubated with the secondary antibodies under 37 °C for 30 min. ECL (Vazyme) was applied to quantitatively label the proteins. Primary antibodies used in this study were CD9 (1:500, Proteintech), CD63 (1:500, Abcam), CD81 (1:500, Proteintech), HSP70 (1:500, CST), Alix (1:500, CST), TSG101 (1:500, Abcam), Calnexin (1:500, CST), β -actin (1:5000, Abclonal), CoL1A1 (1:500, Abclonal), P21 (1:500, Zenbio), 53BP1 (1:500, Zenbio), MMP-1 (1:500, Abclonal), and SIRT3 (1:500, Wanleibio).

Senescence-associated β -galactosidase staining

After the cell treatment was completed, the culture medium was removed and washed 3 times by adding PBS. 1 mL of β -galactosidase staining fixative (CST) was added to each well of a 6-well cell culture plate (ExCell) and fixed for 15 min at room temperature. The β -galactosidase staining fixative was aspirated, and the cells were washed with PBS on a shaker 3 times for 3 min each time. 1 mL of staining working solution was added to each well of a 6-well cell culture plate. The staining working solution was prepared as an example for 6-well cell culture plates: β -galactosidase staining solution A 10 μ L, β -galactosidase staining solution B 10 μ L, β -galactosidase staining solution C 930 μ L, and X-Gal solution 50 μ L. The 6-well plates were sealed with cling film to prevent evaporation and incubated at 37 °C, CO₂-free conditions overnight. Observe under an ordinary light microscope and take pictures.

Intracellular ROS and mitochondrial ROS detection

After completion of cell processing, the culture medium was removed and washed 3 times by adding PBS. The DCFH-DA probe (Beyotime) was diluted with α -MEM at a ratio of 1:1000 to a final concentration of 10 μ mol/L. 1 mL of diluted DCFH-DA probe was added to each well of a 6-well cell culture plate. Incubate at 37 °C in a cell culture incubator for 15 min protected from light. The cells were washed three times with α -MEM to fully remove the DCFH-DA probe that had not entered the cells. The

cells were observed and photographed under the Cell Imaging Microplate Detection System.

Cultivated DFs in 12-well cell culture plates. Prepared MitoSOX Red stock solution and working solution according to the kit instructions (MedChemExpress). The cells were washed 3 times with PBS and 500 μ L of MitoSOX Red working solution was added to each well, and incubated at room temperature for 25 min. Washed with α -MEM on a shaker for 3 times, 5 min each time. Observed under a fluorescence microscope and took photos.

RNA extraction and qRT-PCR

Trizol reagent (Invitrogen) was used to extract total RNA from the cells. The HisyGo RT Red SuperMix for qPCR (+gDNA Wiper) kit (Vazyme) was used to reverse transcribe the RNA samples. Briefly, the amount of RNA was determined based on the concentration of the measured RNA sample, and then 5 $\times$ gDNA Wiper Mix was added and the system was made up to 15 μ L with RNase-free ddH₂O. After mixing, the genomic DNA was removed on a PCR instrument at 42 °C for 2 min. Added 4 $\times$ HisyGo Red SuperMix directly into the reaction tube. cDNA synthesis was completed at 37 °C, 15 min and 85 °C, 5 s on the PCR instrument. Primers, AceQ qPCR SYBR GreenMix (Vazyme), and cDNA samples were sequentially added into tubes according to the instructions. The mRNA expression level of each gene was analyzed by QuantStudio™ 3 real-time fluorescence quantitative PCR instrument (Thermo Fisher Scientific). The results were statistically analyzed using the $2^{-\Delta\Delta C_t}$ method. The primer sequences for the genes are listed in Table S1.

Morphological analysis of mitochondria

Prepared Mito-Tracker Red CMXRos storage solution and working solution according to the kit instructions (Beyotime). The cells were washed with PBS for 3 times. Removed the cell crawler and placed them on slides, added drops of prepared Mito-Tracker Red CMXRos working solution pre-warmed at 37 °C, and incubated for 30 min at 37 °C, protected from light. At the end of the incubation, removed the working solution, and added PBS to wash 3 times. Each well was fixed with 500 μ L of 4% paraformaldehyde fixative for 10 min at room temperature and washed with PBS on a shaking table 3 times, 10 min each time. 500 μ L of 5% BSA was added to each well to block at room temperature for 1 h. The cell crawlers were incubated with diluted Hoechst33342 dye for 10 min at room temperature and then washed with PBS 3 times, 10 min each time. Anti-fluorescence quencher was dropped and then observed under a fluorescence microscope and photographed. Mitochondrial network parameters were calculated by MiNA.

Mitochondrial membrane potential assay

The JC-1 staining working solution and JC-1 staining buffer were prepared according to the kit instructions (Beyotime). For cells cultured in six-well plates, the existing culture medium was removed, and subsequently, 1 mL of fresh complete culture medium was added to each well. Following this, an additional 1 mL of the JC-1 staining working solution should be added to each well, ensuring thorough mixing. The cells should then be incubated in a cell culture incubator for 25 min. At the end of incubation, gently washed with JC-1 staining buffer. Added the complete culture medium, and placed it under the fluorescence microscope for observation and photographic documentation.

Determination of intracellular NAD⁺

Cell lysates, ethanol dehydrogenase, and color development solution were sequentially added to 96-well cell culture plates according to the kit instructions (Beyotime). OD values were measured on a microplate reader (BioTek) at 450 nm.

Statistical analysis

The statistical analysis was carried out using GraphPad Prism 9.0 software (San Diego). The results were all shown as mean $\pm$ standard deviation (SD). To determine differences between the two groups, an unpaired t-test was applied, and for multiple groups, a one-way ANOVA was used. P value < 0.05 was considered statistically significant. * P < 0.05, ** P < 0.01, *** P < 0.001, and **** P < 0.0001.

Results

Nano-microneedles effectively promote sEVs into the deeper layers of the skin and prevent skin aging in mice

The hucMSCs obtained had a typical stem cell shuttle-shaped morphology (Figure S1A), good adipogenic-induced differentiation and osteogenic-induced differentiation abilities (Figure S1B), and expressed the corresponding stem cell positive markers CD105, CD73, and CD166, and low expressed the markers CD34, CD45, and CD11b (Figure S1C). The above results indicated that hucMSCs with multidifferentiation ability were successfully isolated. HucMSC-sEVs obtained from the supernatant exhibited a typical vesicle-like structure (Fig. 1A). The NTA results showed that its particle size was concentrated at 124.2 nm with a zeta potential of -22.66 mV (Fig. 1B-C). HucMSC-sEVs expressed the characteristic markers CD9, CD63, TSG101, HSP70, and Alix, but not the negative marker Calnexin (Fig. 1D). The positivity rates of hucMSC-sEVs markers CD9, CD63, and CD81 by nanoflow assay were 47.8%, 34.4%, and 32.5%, respectively (Fig. 1E).

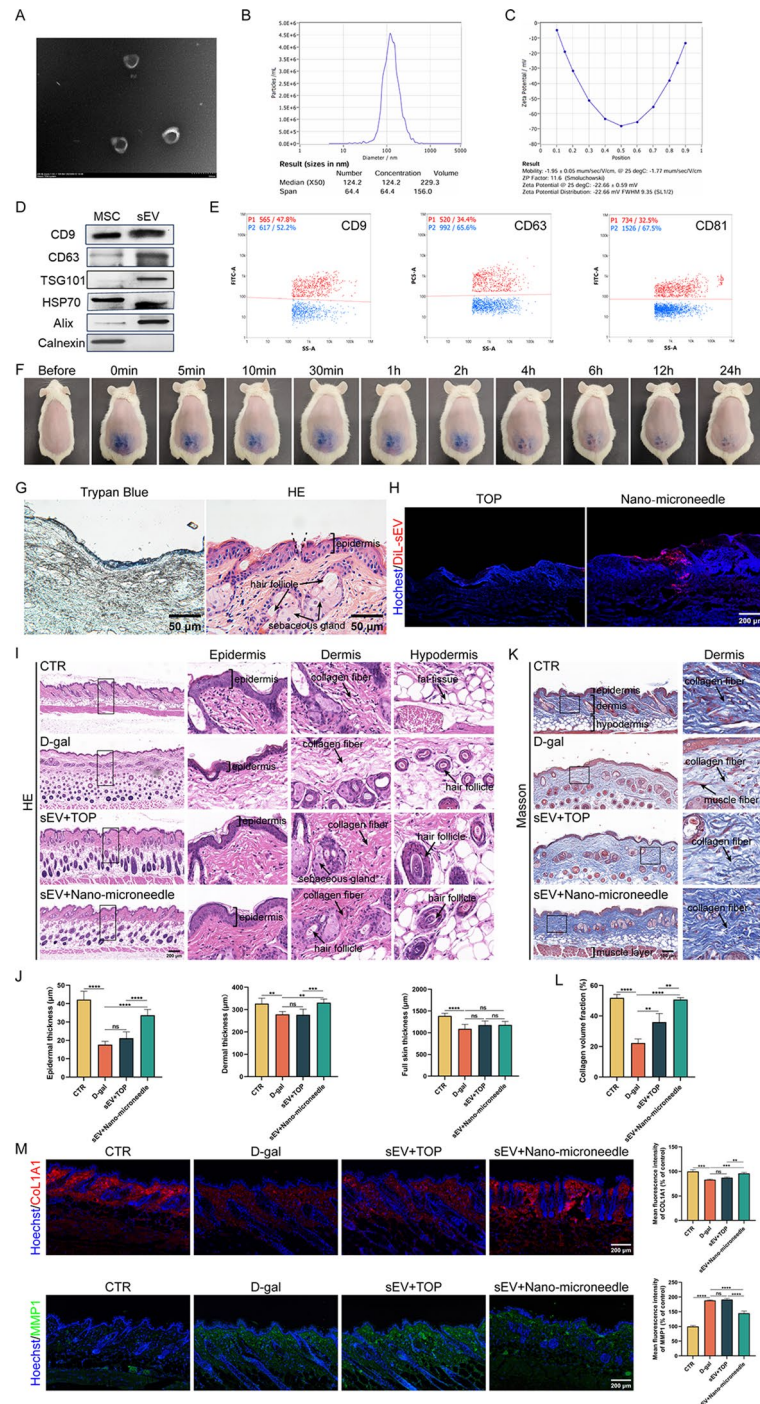


Fig. 1 Nano-microneedles effectively promote sEVs into the deeper layers of the skin and prevent skin aging in mice. **A**. TEM image of hucMSC-sEVs (scale bar = 200 nm); **B**. NTA showed the size distribution of hucMSC-sEVs; **C**. NTA detected the membrane potential of hucMSC-sEVs; **D**. Western blot detected the surface markers of hucMSC-sEVs. Full-length blots are presented in the Additional file: Original Images for Blots; **E**. Positive rates of hucMSC-sEVs markers CD9, CD63, and CD81 detected by nanoflow cytometry; **F**. Skin stratum corneum recovery after nano-microneedles treatment; **G**. Trypan blue staining and HE staining to observe the depth of nano-microneedles penetration (scale bar = 50 µm); **H**. Immunofluorescence to show the distribution of DiL-sEVs in skin tissues after 24 h of smear or nano-microneedles treatment (scale bar = 200 µm); **I**. HE staining of skin tissue (scale bar = 200 µm); **J**. Quantification of skin epidermal layer thickness, quantification of skin dermal layer thickness, and quantification of skin whole layer thickness at the end of the experiment; **K**. Masson staining of skin tissue (scale bar = 100 µm); **L**. Quantification of collagen fiber percentage in skin tissue at the end of the experiment; **M**. Col1A1 and MMP1 expression and their quantification maps (scale bar = 200 µm). Data are presented as mean ± SD. $n = 3$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$, ns: no significant difference

To identify the most effective delivery method for sEVs, we performed a comparative analysis of topical application and nano-microneedles. The recovery of the skin stratum corneum after nano-microneedles action was examined, and it was found that the trypan blue staining started to fade after 10 min of treatment, indicating that the skin barrier function could be recovered soon after nano-microneedles treatment (Fig. 1F). Next, the depth of penetration of the nano-microneedles was examined by applying nano-microneedles to the skin of mice treated with trypan blue. The nano-microneedles could penetrate the epidermis and approach the dermis (Fig. 1G). Immunofluorescence results showed that sEVs could better enter the deeper layers of the skin after 24 h of nano-microneedles administration relative to the topical application (Fig. 1H). HE staining results showed that nano-microneedles administration of sEVs effectively reversed the attenuation of the epidermis and the disarray of collagen fibers in the dermis of senescent mice relative to the topical application (Fig. 1I-J). Masson staining results indicated that the collagen fiber percentage of the skin of senescent mice was significantly reduced, while significantly elevated after nano-microneedles administration of sEVs (Fig. 1K-L). In addition, the expression of Col1A1 and MMP1 was also examined, with decreased expression of Col1A1 and elevated expression of MMP1 in senescent mice, which was effectively reversed by nano-microneedles administration of sEVs, relative to topical application (Fig. 1M). Therefore, for subsequent experiments, we used nano-microneedles as a delivery method.

NMN-loaded hucMSC-sEVs effectively delay skin aging in mice

The anti-aging impact of NMN, as a representative of anti-aging drugs that modulate mitochondrial function, has been demonstrated in preclinical animal studies and human clinical trials [51]. However, the anti-skin aging effect of NMN remains controversial due to its high water solubility and low permeability. We constructed NMN-sEVs using electroporation and validated their loading efficiency. The sEVs with different concentrations (1×10^{10} particles/mL, 5×10^{10} particles/mL, and 1×10^{11} particles/mL) were treated with 30 mM NMN by electroporation. The TEM results showed that NMN-sEVs have a typical cup-shaped structure (Fig. 2A). In addition, the NTA results indicated that the particle size of NMN-sEVs was concentrated around 140 nm and had a zeta potential of around -20 mV (Fig. 2B-C). The particle size and the absolute value of zeta potential of NMN-sEVs have no significant difference from natural sEVs. To determine whether the NMN-sEVs carried NMN, we applied HPLC to examine the loading efficiency (Figure S2). The results showed that the encapsulation rate could reach up to

$44.13 \pm 4.07\%$, and the drug loading capacity could reach up to 4424.57 ± 407.66 $\mu\text{g}/10^{10}$ sEV when the concentration of the sEVs was 1×10^{10} particles/mL (Table 1; Fig. 2D). As the concentration of hucMSC-sEVs increased, the encapsulation rate and drug loading gradually decreased. The above results indicated that we successfully prepared NMN-sEVs by applying the electroporation technique. Different concentrations of NMN (10 mM, 30 mM, and 50 mM) were electroporated with 1×10^{10} particles/mL hucMSC-sEVs. β -galactosidase staining and ROS results showed that NMN-sEVs obtained by electroporation with different concentrations of NMN were effective in ameliorating the increase in the number of senescent cells and the elevation of the oxidative stress level induced by D-gal (Figure S3). However, it is noteworthy that the effect was not further enhanced when the NMN concentration was 50 mM (Figure S3). Based on the above results, in future experiments, we will select 1×10^{10} particles/mL hucMSC-sEVs and 30 mM NMN to construct NMN-sEVs, which have the highest encapsulation efficiency and drug loading capacity, as well as the best effect.

To observe the absorption and metabolizability of sEVs and NMN-sEVs in mice, we applied DiR-labeled sEVs and NMN-sEVs to the skin via nano-microneedles and found that the delivered sEVs and NMN-sEVs were metabolized at an accelerated rate only after 72 h (Fig. 2E), indicating that sEVs and NMN-sEVs could remain active and functional in the skin tissue for a period of time. To assess the potential beneficial impact of NMN-sEVs on skin aging, a D-gal-induced skin aging model was created. Through the gross view of skin tissues, we observed that fine wrinkles were clearly visible on the back skin of mice in the D-gal group; however, the mice in the PBS group saw no improvement in their wrinkles. In contrast, the depth of wrinkled folds was attenuated on the backs of mice in the sEV and NMN groups; notably, the skin condition of the mice treated with NMN-sEVs was almost indistinguishable from that of the normal state (Fig. 2F). The results of HE staining indicated that after the long-term application of D-gal, the epidermal layer of mice thinned, and the sEV, NMN, and NMN-sEV groups all improved, with the most significant improvement in the NMN-sEV group (Fig. 2G). Masson staining revealed a significant reduction in the percentage of collagen fibers in the D-gal group of mice compared to the control group. Conversely, mice in the NMN-sEV group exhibited a notable enhancement in collagen fiber levels (Fig. 2H). In addition, immunofluorescence and immunohistochemical staining revealed that the levels of aging-related biomarkers P21 and P53 increased, the expression of proliferation indicators ki67 and PCNA decreased, and the expression of DNA damage signals γ -H2AX and 53BP1 increased in the skin tissue of D-gal group mice (Fig. 2I-J). Intervention with NMN-sEVs significantly improved oxidative stress damage and inhibited

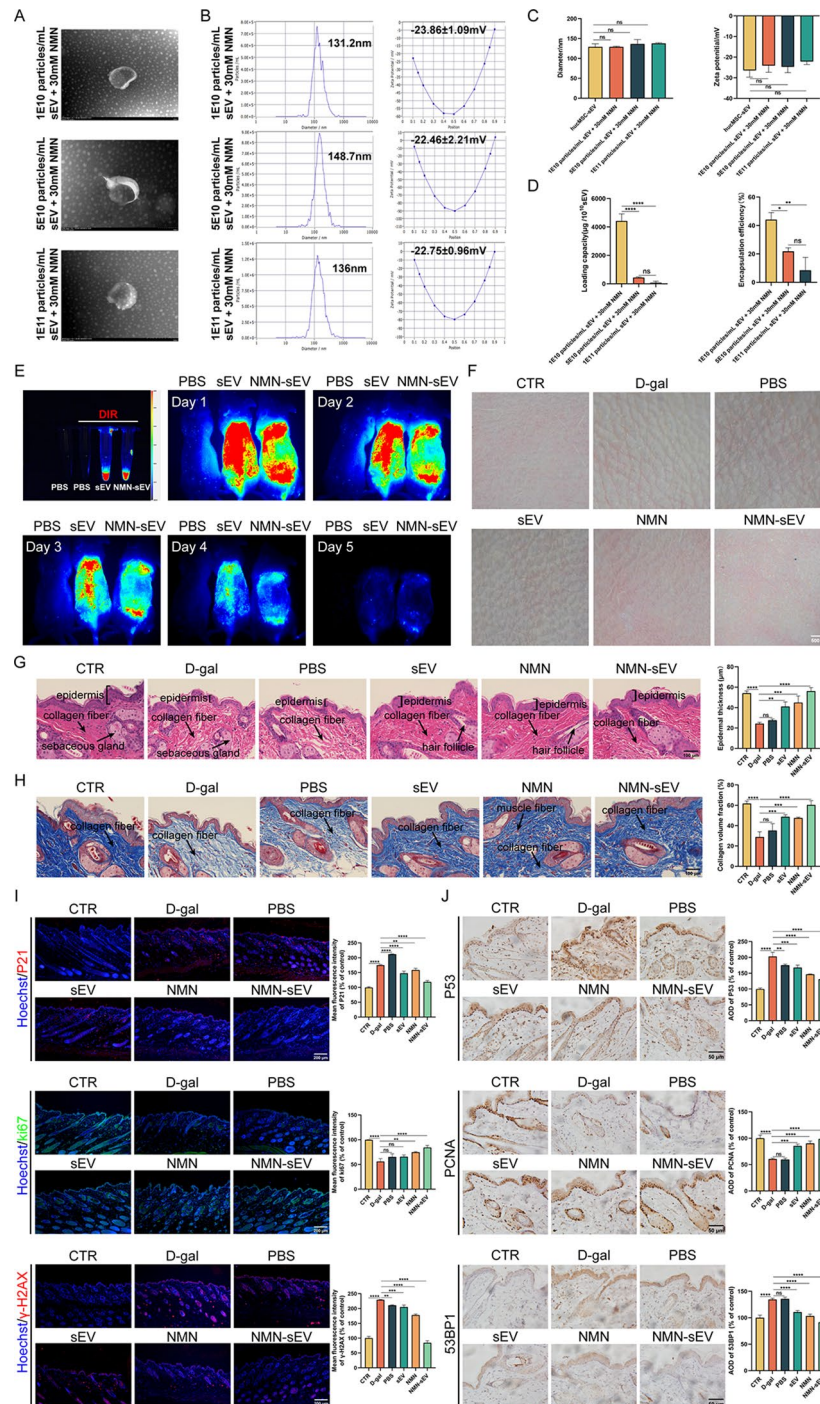


Fig. 2 NMN-sEVs effectively delay skin aging in mice. **A**. TEM detection of NMN-sEVs morphology (scale bar = 200 nm); **B-C**. NTA detected the particle size and zeta potential of NMN-sEVs; **D**. HPLC detection of the encapsulation rate and loading capacity of NMN at different concentrations of sEVs; **E**. Skin metabolism in mice after application of sEVs or NMN-sEVs assisted by nano-microneedles; **F**. Gross view of mice dorsal skin (scale bar = 500 μm); **G**. HE staining of skin tissues (scale bar = 100 μm) and quantification of epidermal layer thickness; **H**. Masson staining of skin tissue (scale bar = 100 μm) and quantification of collagen fiber percentage in skin tissue; **I**. Tissue immunofluorescence to detect the levels of P21, ki67, and γ-H2AX and their quantification maps (scale bar = 200 μm); **J**. Tissue immunochemical staining to detect the expression of P53, PCNA, and 53BP1 and their quantification maps (scale bar = 50 μm). Data are presented as mean ± SD. $n = 3$, $P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$, ns: no significant difference

Table 1 Encapsulation efficiency and loading capacity of NMN at different concentrations of sEVs

Sample	Encapsulation Efficiency M ± SD (%)	Loading Capacity M ± SD (μg/10 ¹⁰ sEV)
1 × 10 ¹⁰ particles/mL sEVs + 30 mM NMN	44.13 ± 4.07	4424.57 ± 407.66
5 × 10 ¹⁰ particles/mL sEVs + 30 mM NMN	21.82 ± 1.77	437.53 ± 35.46
1 × 10 ¹¹ particles/mL sEVs + 30 mM NMN	8.46 ± 7.42	84.81 ± 74.36

proliferation ability in aging mice, confirming that NMN-sEVs were effective in delaying skin aging in mice. To confirm the safety of NMN-sEVs, we also examined the body weight (Figure S4A), serum biochemical indexes (Figure S4B), and the structure of major organs (Figure S4C) of the mice and found that there were no significant changes, which proved the safety of NMN-sEVs.

NMN-loaded hucMSC-sEVs effectively defend against DFs senescence

To further assess whether NMN-sEVs could attenuate cellular senescence and restore cellular function, we established a model of DFs senescence induced by D-gal. DFs isolated from the dorsal skin of SD rats grew in a swirling pattern around the skin tissue mass, and individual cells had a short shuttle shape (Figure S5A). Cellular immunofluorescence (Figure S5B) and western blot (Figure S5C) showed that the isolated DFs expressed vimentin and α-SMA and did not express CK10, which is consistent with DF characteristics. DFs were subjected to varying concentrations of D-gal in a dose-dependent manner, and the number of live DFs gradually decreased with increasing D-gal concentration, with the vast majority of the DFs dying at 80 mg/mL (Figure S5D). In addition, when the D-gal concentration was 10 mg/mL, the number of senescent cells began to increase significantly, and the level of ROS also began to increase (Figure S5E-F). Meanwhile, with the increase of D-gal concentration, the proliferative vigor of DFs gradually decreased (Figure S5G); the levels of senescence-related markers MMP1, P16, and P53 was elevated, and the expression of Col1A1 was decreased (Figure S5H-I); and the number of cells deficient in LaminB1 gradually increased (Figure S5J). Taking these results together, 10 mg/mL D-gal (most cells senesced without significant changes in cell number) was selected as the subsequent treatment concentration.

To assess the uptake of sEVs and NMN-sEVs by DFs, DiI-labeled sEVs and NMN-sEVs were co-incubated with DFs for 24 h. The results showed that DiI-labeled sEVs and NMN-sEVs could be efficiently uptaken by DFs (Fig. 3A). The cell proliferation viability was detected after co-incubation of sEVs, NMN, and NMN-sEVs

with DFs, with no significant changes, indicating that sEVs, NMN, and NMN-sEVs had no toxic effects on DFs (Fig. 3B). The sEVs, NMN, and NMN-sEVs were effective in rescuing the reduction of proliferative viability of DFs caused by D-gal, and the effect of NMN-sEVs was the most significant (Fig. 3C). In addition, the addition of sEVs, NMN, or NMN-sEVs was effective in reducing the level of ROS (Fig. 3D) and β-galactosidase activity (Fig. 3E) in DFs, with the NMN-sEVs intervention group exhibiting the most pronounced effect. Similarly, the anti-aging effect was found to be more significant in the NMN-sEV group compared with the sEV and NMN groups by detecting the levels of *Il-6*, *Il-8*, and *Tgf-beta*, as well as the senescence-related markers *Col1A1*, *53BP1*, and *P21* (Fig. 3F-G). The cellular immunofluorescence results also confirmed that NMN-sEVs intervention was effective in attenuating D-gal-induced cell cycle block and DNA damage and down-regulating the expression of the cell cycle blocking molecule P16 and the DNA double-strand break marker γ-H2AX (Fig. 3H). To examine cell proliferation, we also performed ki67 staining, which showed that NMN-sEVs better promoted cell proliferation compared with sEVs and NMN (Fig. 3H). In conclusion, the above results suggest that NMN-sEVs may be more effective in safeguarding DFs from oxidative stress and cellular senescence induced by D-gal compared with sEVs and NMN.

NMN-loaded hucMSC-sEVs ameliorate cellular senescence by promoting mitochondrial autophagy

Mitochondrial dysfunction represents a significant characteristic of cellular senescence. Therefore, we further performed mitochondrial electron microscopy to explore whether NMN-sEVs delayed skin aging by improving mitochondrial function. The findings from mitochondrial electron microscopy indicated that the mitochondrial cristae structures of senescent cells were blurred or disappeared, while the mitochondrial morphology of NMN-sEVs-intervened cells improved and autophagy appeared, so we further speculated that NMN-sEVs might delay senescence by regulating mitochondrial autophagy (Fig. 4A). We detected mitochondrial morphology by Mitotracker; normal mitochondria were filamentous and rod-shaped, while mitochondria in senescent cells fragmented into dots, which was significantly improved by NMN-sEVs (Fig. 4B). The results of mitochondrial membrane potential (Fig. 4C) and mitochondrial ROS (Fig. 4D) indicated a reduction in mitochondrial membrane potential and an elevation in ROS levels in senescent cells, which was reversed by NMN-sEVs, confirming that NMN-sEVs improved the mitochondrial quality and function. In addition, mitochondrial co-localization with LC3B and mitochondrial co-localization with LAMP1 (indicator lysosome) were inhibited in senescent

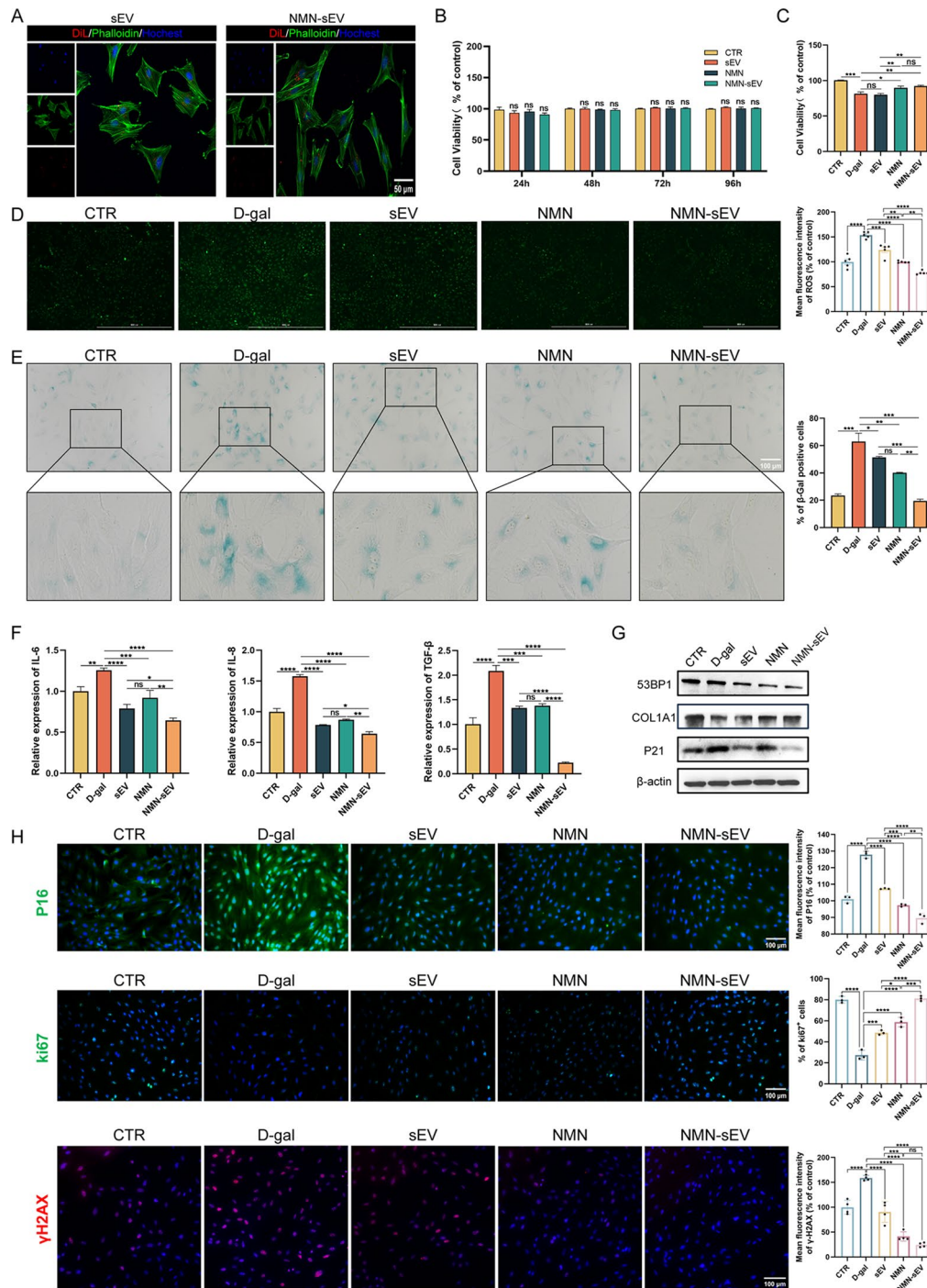


Fig. 3 NMN-sEVs effectively defend against DFs senescence. **A**. Fluorograms of DiI-labeled sEVs and NMN-sEVs co-incubated with DFs (scale bar = 50 μ m); **B**. Changes in cell proliferative capacity of DFs after treatment with sEVs, NMN, and NMN-sEVs; **C**. Assay of cell proliferative viability; **D**. Fluorescence intensity of ROS and its quantification graph (scale bar = 1000 μ m); **E**. β -galactosidase activity assay and its quantification graph (scale bar = 100 μ m); **F**. Quantification of *Il-6*, *Il-8*, and *Tgf-beta*; **G**. Changes in senescence-related markers in DFs detected by Western Blot. Full-length blots are presented in the Additional file: Original Images for Blots; **H**. Cellular immunofluorescence assay for P16, ki67, and γ -H2AX expression and their quantification graphs (scale bar = 100 μ m). Data are presented as mean $\pm$ SD. $n = 3$, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: no significant difference

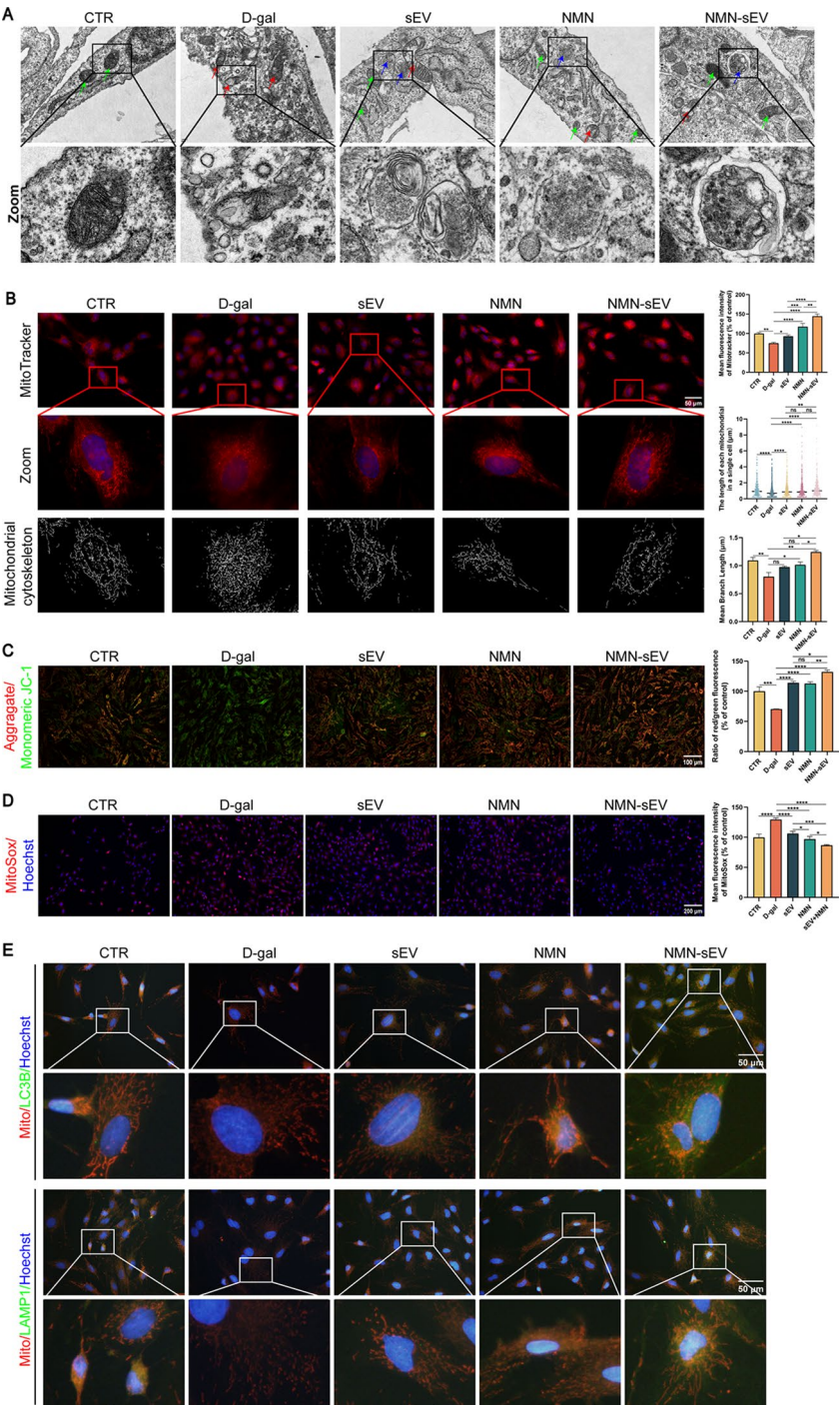


Fig. 4 NMN-sEVs ameliorate cellular senescence by promoting mitochondrial autophagy. **A**. Representative TEM images of mitochondria from differently treated DFs. The green arrow represents normal mitochondria, the red arrow represents damaged mitochondria, and the blue arrow represents autophagic lysosomes (scale bar = 500 nm); **B**. Mitotracker detection of mitochondrial morphology (scale bar = 100 μ m); **C**. Representative fluorescence microscopy images of JC-1 and its quantification map (scale bar = 100 μ m); **D**. Mitochondrial ROS fluorescence intensity and its quantification map (scale bar = 200 μ m); **E**. Images of co-localization of mitochondria with LC3B and LAMP1 (indicator lysosome) respectively (scale bar = 50 μ m). Data are presented as mean $\pm$ SD. $n = 3$, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: no significant difference

fibroblasts and increased after NMN-sEVs treatment (Fig. 4E). These results suggest a defect in mitochondrial autophagy in senescent cells and a significant positive effect of NMN-sEVs on mitochondrial autophagy.

NMN-loaded hucMSC-sEVs modulate NAD⁺/SIRT3 pathway-mediated mitochondrial autophagy to delay DFs aging

To further explore the specific mechanism by which NMN-sEVs promote mitochondrial autophagy, we screened aging-related genes associated with mitochondrial dysfunction from the Aging Atlas database (Table S2). We found that SIRT3, a NAD⁺-dependent deacetylase, was reported to be associated with mitochondrial autophagy. NAD⁺ levels and autophagic activity decline with aging, leading to impaired communication between mitochondria and the nucleus, as well as the accumulation of damaged organelles, including mitochondria. In contrast, mitochondrial autophagy-induced mitochondrial elimination ameliorates cellular senescence and inhibits the secretion of SASP factors. By supplementing NMN, the precursor of NAD⁺, autophagy and mitochondrial quality control could be enhanced, thereby promoting tissue rejuvenation and organismal lifespan extension. Therefore, we speculated that NMN-sEVs might delay cellular senescence by promoting NAD⁺/SIRT3 pathway-mediated mitochondrial autophagy. To prove this conjecture, we found an interaction between SIRT1, a homologous family member of SIRT3, and NAD⁺ by screening proteins interacting with NAD⁺ in the STITCH database (Fig. 5A). Therefore, we first verified the interactions between SIRT3 and NAD⁺ by molecular docking (Fig. 5B). In addition, we examined the NAD⁺ level in DFs and found that the NAD⁺ level in DFs was reduced after D-gal induction, while NMN-sEVs significantly restored the intracellular NAD⁺ level (Fig. 5C). Cellular immunofluorescence and immunohistochemical staining showed that SIRT3 was localized on mitochondria and that NMN-sEVs could promote the expression of SIRT3 more efficiently in DFs (Fig. 5D) and skin tissues (Fig. 5E) compared to sEVs and NMN. More importantly, Western blot detection of the expression of SIRT3 and mitochondrial autophagy-related proteins, including mTOR, LC3, Parkin, PINK1, and P62, and the results revealed that the expression of SIRT3 was increased and mitochondrial autophagy was activated by pretreatment with NMN-sEVs (Fig. 5F), suggesting that NMN-sEVs might regulate NAD⁺/SIRT3 pathway-mediated mitochondrial autophagy to delay aging.

To further confirm the beneficial role of SIRT3, an effector molecule up-regulated after NMN-sEVs intervention, in delaying cellular senescence, we used 3-TYP, a selective inhibitor of SIRT3. DFs were treated with different concentrations of 3-TYP, and 100 μ M, which did not

induce strong cytotoxicity, was identified as the subsequent concentration of 3-TYP to be used (Fig. 5G). Consistent with previous findings, NMN-sEVs improved the proliferative viability of senescent cells, whereas 3-TYP treatment inhibited the improvement of cell proliferative viability by NMN-sEVs (Fig. 5H). Meanwhile, 3-TYP treatment attenuated NMN-sEVs-induced inhibition of β -galactosidase activity, suggesting that the SIRT3-selective inhibitor, 3-TYP, inhibited the improvement of cellular senescence by NMN-sEVs (Fig. 5I). To further demonstrate that the cellular senescence-delaying effect of NMN-sEVs correlates with mitochondrial autophagy mediated by the NAD⁺/SIRT3 pathway, we assessed the effect of 3-TYP treatment on mitochondrial autophagy. Western blot detected the expression of mitochondrial autophagy-related proteins, including Parkin, PINK1, P62, p-mTOR, and mTOR after 3-TYP treatment, and the results showed that mitochondrial autophagy was inhibited by 3-TYP pretreatment (Fig. 5J). The mitochondrial ROS results showed that NMN-sEVs reduced mitochondrial ROS levels in senescent cells, whereas 3-TYP treatment could again elevate the level of oxidative stress in cells (Fig. 5K). Furthermore, mitochondrial membrane potential was elevated after NMN-sEVs intervention and decreased after 3-TYP treatment (Fig. 5L). More importantly, mitochondrial distribution in DFs became dispersed and fragmented after 3-TYP treatment, and co-localization of mitochondria and LC3B, as well as mitochondria and LAMP1, was attenuated, similar to D-gal-induced senescent DFs, suggesting that the activation of mitochondrial autophagy by NMN-sEVs was inhibited by 3-TYP (Fig. 5M). The above results confirmed that SIRT3 plays an important role in the activation of mitochondrial autophagy by NMN-sEVs to retard skin aging.

Discussion

Aging represents the most significant risk factor for numerous diseases and is closely related to progressive functional defects at the cellular, tissue, and organismal levels. The skin is one of the first organs to show signs of aging. Currently, the doctrine of skin aging includes cellular senescence, gene regulation, free radical damage, metabolic changes, glycosylation, immune dysregulation, and hormonal dysregulation [52]. Among them, AGEs are an important factor leading to skin aging, which are produced by the carbonyl groups of proteins and reducing sugars in the body under non-enzymatic conditions. On one side, they attach to the receptors of specific cells, affecting the expression of a variety of inflammatory cytokines, leading to inflammatory necrosis of skin cells, and ultimately aging or even cancer; on the other side, AGEs can also bind to proteins, affecting their normal structure and function, resulting in a decrease in elasticity

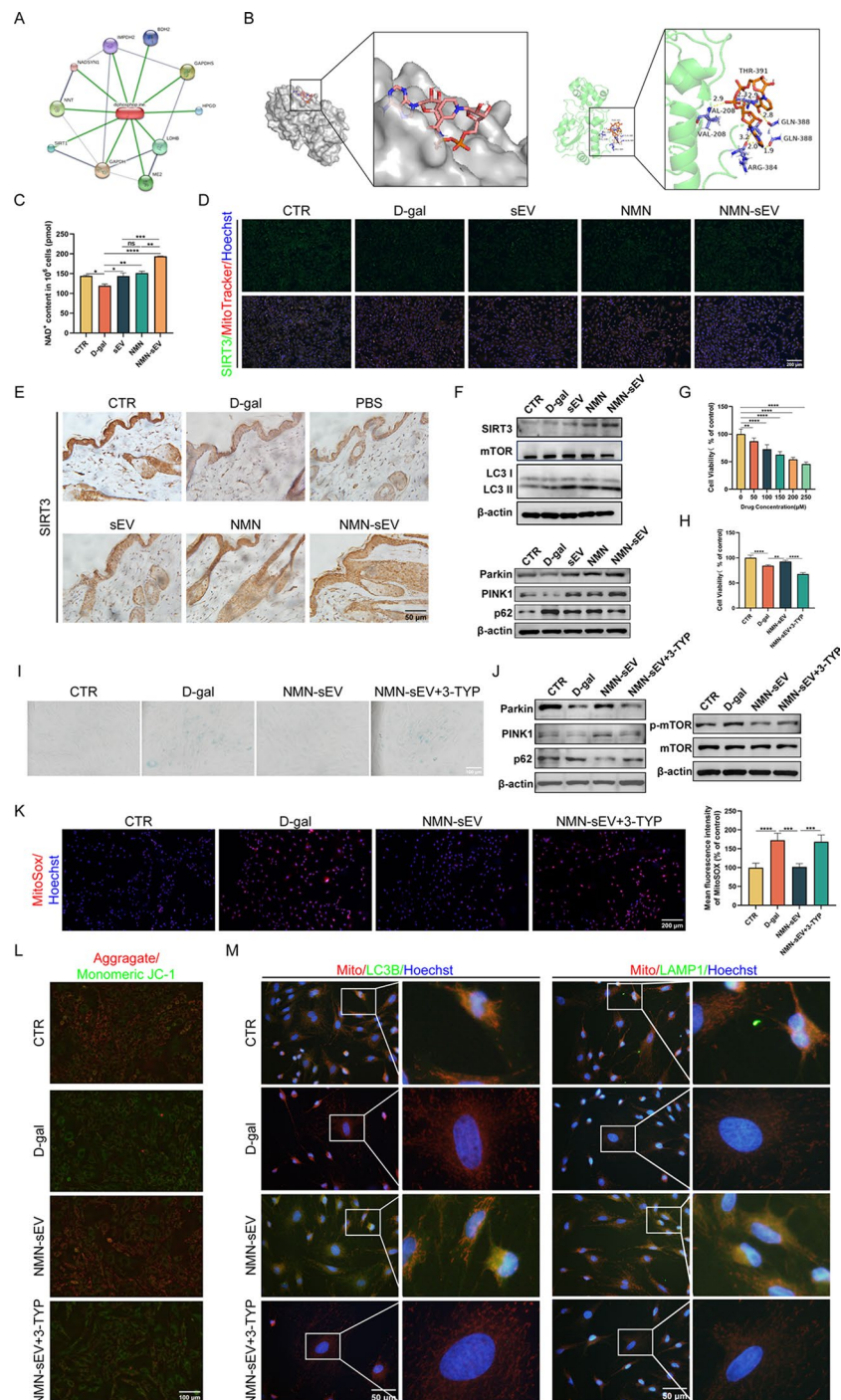


Fig. 5 NMN-sEVs modulate NAD⁺/SIRT3 pathway-mediated mitochondrial autophagy to delay DF aging. **A**. STITCH database screening for NAD⁺ interacting proteins; **B**. Molecular docking results of NAD⁺ and SIRT3; **C**. NAD⁺ content in DFs; **D**. SIRT3 expression in DFs (scale bar = 200 μm); **E**. SIRT3 expression in skin tissues (scale bar = 50 μm); **F**. Western blot examined SIRT3, mTOR, LC3, Parkin, PINK1, and P62 expression. Full-length blots are presented in Additional file: Original Images for Blots; **G**. Cell proliferation viability assay of DFs after treatment with different concentrations of 3-TYP for 24 h; **H**. Cell proliferation viability assay of DFs after different treatments; **I**. β-galactosidase staining assay (scale bar = 100 μm); **J**. Western blot examined Parkin, PINK1, P62, p-mTOR, and mTOR expression. Full-length blots are presented in the Additional file: Original Images for Blots; **K**. Fluorescence intensity of mitochondrial ROS and its quantification map (scale bar = 200 μm); **L**. Representative fluorescence microscopy images of co-localization of mitochondria with LC3B as well as LAMP1 (indicator lysosome), respectively (scale bar = 50 μm). Data are presented as mean ± SD. *n* = 3, **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001, ns: no significant difference

of the skin, deepening of wrinkles, and accelerating the aging process of the skin [53, 54]. D-gal is a powerful glycosylation agent that promotes the production of AGEs and elevated levels of oxidative stress in the organism. The D-gal animal model of aging has become one of the most recognized animal models of the aging state and is widely used in anti-aging research [55, 56]. Consistent with previous studies, we found here that D-gal resulted in a deepening of wrinkles and skin thinning in mice and elevated levels of oxidative stress in DFs. As the most external barrier of the human body, the skin interacts and communicates deeply with the organism. Research has demonstrated that skin aging is linked to various factors, including cellular senescence phenotypes, immune senescence, longevity, and cardiovascular disease risk. In addition, skin aging induced by UV radiation affects the brain by reducing hippocampal neurogenesis and activating the central hypothalamic-pituitary-adrenal axis [11]. Therefore, topical application of anti-skin aging drugs may play a significant role in the formulation of novel anti-aging strategies, potentially postponing the onset of systemic aging and age-related diseases.

MSCs can be isolated from a wide variety of human tissues and organs, including adipose tissue, umbilical cord, and bone marrow, and have shown promising therapeutic potential in regenerative dermatology, such as anti-scarring, hair regeneration, promotion of wound healing, and rejuvenation of the skin, through paracrine mechanisms (e.g., secretion of EVs) [57–59]. For instance, Zhang et al. [60] discovered that hucMSC-EVs could effectively alleviate skin aging by reducing UVB-induced Notch activation through the key protein component TIMP1 in their cargo. Nevertheless, the effect of natural EVs in delaying skin aging is limited, and engineering EVs to enhance their therapeutic potential is a future trend. Therefore, we intend to construct an engineered EV for specific therapeutic use. Given the promising therapeutic potential of NMN in attenuating cellular senescence, restoring mitochondrial function, and prolonging lifespan, as well as its low permeability and instability in direct drug delivery, the modification of EVs with NMN presents a viable strategy to leverage the benefits of both. Here, we constructed NMN-sEVs by electroporation and verified their loading efficiency by HPLC. Relative to higher concentrations, lower concentrations of sEVs encapsulated NMN more efficiently. We speculated that it might be because higher concentrations of sEVs are more prone to aggregation under the action of the electric field rather than loading NMN. Regarding the application of NMN-sEVs in the skin, their absorption rate is limited due to the skin barrier. Nano-microneedles are emerging permeation tools made of high-purity monocrystalline silicon with a diameter of about one-thousandth of a hair's diameter, which can form open drug delivery channels

on the skin's surface in a short period to promote transdermal penetration and absorption of topical drugs [61, 62]. Moreover, the channels they form can close on their own in a short period, with a low possibility of infection. Compared with topical application and intradermal injection, nano-microneedles are more efficient in drug delivery and less likely to cause skin irritation and inflammation. Consistently, our findings suggested that nano-microneedles have advantages over topical application in effectively facilitating the entry of sEVs into the deeper layers of the skin and reducing skin thinning and collagen loss due to aging. NMN-sEVs were more effective in improving aging-induced wrinkles, skin thinning, reduced collagen fiber percentage, proliferation inhibition, cellular senescence, and oxidative damage than sEVs or NMN alone.

Characteristics of senescent cells include DNA damage, metabolic changes, endoplasmic reticulum stress, and mitochondrial dysfunction [63, 64]. Mitochondria are crucial for modulating age-associated decline, and mitochondrial dysfunction induces cellular aging and the secretion of SASP [65]. In the context, we examined mitochondrial function in NMN-sEVs-treated DFs and found that NMN-sEVs rescued senescence-induced mitochondrial dysfunction, which was characterized by a compromised mitochondrial structure, low membrane potential, and high levels of ROS. We further found the activation of mitochondrial autophagy, the elevated level of NAD⁺, and the increase in the expression of SIRT3 in the NMN-sEVs treatment. NAD⁺ serves as a crucial coenzyme in cellular energy metabolism and may represent an adaptive mechanism in response to oxidative stress, participating in metabolic pathways and influencing mitochondrial function [66]. SIRT3 is a NAD⁺-dependent deacetylase located on mitochondria that regulates mitochondrial autophagy through metabolic rewiring and NAD⁺ depletion [38]. In addition, SIRT3 has the capacity to augment the activity of intrinsic antioxidant enzymes and inhibit the generation of ROS, thus safeguarding cells against oxidative injury [67, 68]. NAD⁺ depletion reduces the expression of SIRT3 and increases the acetylation level of its target proteins. As NMN-sEVs supplementation enhances NAD⁺ levels, NMN-sEVs may activate SIRT3 expression by promoting NAD⁺ synthesis to regulate mitochondrial autophagy. Therefore, we used 3-TYP to verify the regulatory effect of SIRT3 on mitochondrial autophagy. The findings indicated that 3-TYP notably hindered the improvement of cellular senescence, aberrant mitochondrial function, and mitochondrial autophagy inhibition by NMN-sEVs and restored the senescence-associated phenotypic features, confirming that SIRT3 is a key molecule in the regulation of mitochondrial autophagy in the process of senescence. In addition to mitochondria, the structure of

endoplasmic reticulum (ER) was also affected by senescence. Our results indicated that D-gal induces endoplasmic reticulum expansion and degranulation in DFs, while the expanded endoplasmic reticulum in NMN-sEV group is significantly reduced (Figure S6). There are mitochondria-ER contacts (MERCs) between the ER and mitochondria, and frequent exchange of substances, including Ca^{2+} and lipids, also occurs [69]. In addition, mitochondria, ER, and MERCs are involved in ROS production and autophagosome assembly. As the number of cell passages increases, mitochondrial calcium uptake decreases, and the number of MERCs decreases in senescent cells, supporting the potential of MERCs as a therapeutic target for senescence that may affect cellular function and lifespan [70]. In the future, we will further investigate how NMN-sEVs regulate the structure and function of MERCs and their roles in anti-aging, providing new ideas and targets for anti-aging research.

There are several limitations to this study. It is essential to comprehend the molecular alterations in engineered EVs and how NMN-sEVs interact with and influence the behavior of senescent cells during the regeneration of aging skin. Here, we found that mitochondrial dysfunction increased with skin aging, including abnormal mitochondrial morphology and structure, elevated ROS levels, mitochondrial membrane potential loss, and reduced levels of mitochondrial autophagy. Our data then showed that NMN-sEVs significantly delayed skin aging, such as improving skin wrinkles in aging mice in vivo as well as reducing oxidative stress levels in DFs in vitro. In particular, NMN-sEVs increased intracellular NAD^+ levels and SIRT3 expression, as well as rescued the inhibition of senescence-induced mitochondrial autophagy, suggesting a role for NMN-sEVs in delaying skin senescence through NAD^+ /SIRT3 activation of mitochondrial autophagy, which may be a potential target for delaying skin senescence. We demonstrated the role of NMN-sEVs in regulating mitochondrial autophagy in skin aging, but its specific regulatory mechanism still needs further study.

Conclusion

Our study confirmed that senescent cells show mitochondrial dysfunction, including abnormal mitochondrial morphology and structure, elevated ROS levels, loss of mitochondrial membrane potential, and reduced levels of mitochondrial autophagy. NMN-sEVs could rescue senescence-induced dysfunctional mitochondrial accumulation and inhibition of mitochondrial autophagy and attenuate skin senescence through activation of the NAD^+ /SIRT3 signaling pathway, suggesting a role for NMN-sEVs in remodeling mitochondrial function through mitochondrial autophagy. Our results provide theoretical justification for the utilization of NMN-sEVs

in anti-skin aging. In the future, we will investigate more deeply whether NMN-sEVs can be used as an anti-aging drug and further explore the specific mechanism by which NMN-sEVs regulate aging through the activation of mitochondrial autophagy by NAD^+ /SIRT3.

Abbreviations

HucMSC	Human umbilical cord mesenchymal stromal cell
sEVs	Small extracellular vesicles
NMN	β -Nicotinamide mononucleotide
NMN-sEVs	NMN-loaded hucMSC-sEVs
UV	Ultraviolet
D-gal	D-galactose
AGEs	Advanced glycation end products
SASPs	Senescence-associated secretory phenotypes
EVs	Extracellular vesicles
MMPs	Matrix metalloproteinases
ROS	Reactive oxygen species
SIRT3	Sirtuin family deacetylases
MSCs	Mesenchymal stromal cells
IEVs	Large EVs
α -MEM	Minimum essential medium α
FBS	Fetal bovine serum
DFs	Dermal fibroblasts
PBS	Phosphate-buffered saline
TEM	Transmission electron microscopy
NTA	Nanoparticle tracking analyzer
IHC	Immunohistochemical
ER	Endoplasmic reticulum
MERCs	Mitochondria-ER contacts

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13287-025-04460-w>.

Supplementary Material 1

Supplementary Material 2

Acknowledgements

The authors declare that they have not use AI-generated work in this manuscript in this section.

Author contributions

Zixuan Sun, Jiali Li, and Yuzhou Zheng contributed equally to this work. Yulin Tan and Hui Qian conceptualized and reviewed the paper. Zixuan Sun, Jiali Li, and Yuzhou Zheng drafted the manuscript and prepared the figures and tables. Jiaxin Zhang, Wenhuan Bai, Xinyi Deng, Zhijing Wu, XueZhong Xu, and Wei Ding assisted the manuscript preparation. All authors reviewed and approved the final manuscript.

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Data availability

All data in this study are available from the corresponding author upon reasonable request. Mitochondrial dysfunction-related senescence-associated genes are publicly available through the Aging Atlas database (<https://ngdc.cncb.ac.cn/aging/index>). 3D structures of small molecules are publicly available through the pubchem database (<https://pubchem.ncbi.nlm.nih.gov/>). The 3D structures of proteins are publicly available through the PDB database (<https://www.rcsb.org/>).

Declarations

Ethical approval and consent to participate

The Ethics Committee of the Fourth Affiliated Hospital of Jiangsu University approved the experimental protocol for the study of umbilical cord mesenchymal stem cells and exosomes in delaying skin aging (Project title: Study of the roles of umbilical cord mesenchymal stem cells and exosomes in the repair of tissue injuries; Approval number: 202301; Approval date: January 4, 2023). The Fourth Affiliated Hospital of Jiangsu University confirmed that there was initial ethical approval for collection of hucMSCs, and the donors had signed informed consent. All experiments involving animals were approved by the Institutional Animal Care and Use Committee of Jiangsu University and were conducted in accordance with the ARRIVE guideline 2.0 (Project title: NMN-engineered hucMSC-sEV delays D-gal-induced skin aging in mice; Approval number: UJS-IACUC-2024070902; Approval date: July 9, 2024).

Consent for publication

Not applicable.

Competing interests

The authors confirm that there are no conflicts of interest regarding the publication of this article.

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References

- Papaccio F, A DA, Caputo S, Bellei B. Focus on the contribution of oxidative stress in skin aging. *Antioxid (Basel)*. 2022;11(6).
- Wang Z, Wang Q, Wang L, Xu W, He Y, Li Y, et al. Improvement of skin condition by oral administration of collagen hydrolysates in chronologically aged mice. *J Sci Food Agric*. 2017;97(9):2721–6.
- Fuchs E. Scratching the surface of skin development. *Nature*. 2007;445(7130):834–42.
- Wang L, Jiang Y, Wang XX, Zhou J, Cui HY, Xu WD, et al. Effect of oral administration of collagen hydrolysates from Nile tilapia on the chronologically aged skin. *J Funct Foods*. 2018;44:112–7.
- Wang L, Zhang Y, Zhu ZL, Zheng FP, Gao RC. Food-derived collagen peptides: safety, metabolism, and anti-skin-aging effects. *Curr Opin Food Sci*. 2023;51.
- Li CY, Fu Y, Dai HJ, Wang Q, Gao RC, Zhang YH. Recent progress in preventive effect of collagen peptides on Photoaging skin and action mechanism. *Food Sci Hum Well*. 2022;11(2):218–29.
- Pageon H, Zucchi H, Rousset F, Monnier VM, Asselineau D. Skin aging by glycation: lessons from the reconstructed skin model. *Clin Chem Lab Med*. 2014;52(1):169–74.
- Wang H, Wei S, Xue X, You Y, Ma Q. Adipose stem cells' antagonism in glycosylation of D-galactose-induced skin aging of nude mice and its skin recovery function. *Int J Immunopathol Pharmacol*. 2016;29(3):376–85.
- Zhou Y, Xu Q, Dong Y, Zhu S, Song S, Sun S. Supplementation of mussel peptides reduces aging phenotype, lipid deposition and oxidative stress in D-Galactose-Induce aging mice. *J Nutr Health Aging*. 2017;21(10):1314–20.
- Han M, Ban JJ, Bae JS, Shin CY, Lee DH, Chung JH. UV irradiation to mouse skin decreases hippocampal neurogenesis and synaptic protein expression via HPA axis activation. *Sci Rep*. 2017;7(1):15574.
- Franco AC, Aveleira C, Cavadas C. Skin senescence: mechanisms and impact on whole-body aging. *Trends Mol Med*. 2022;28(2):97–109.
- Salotti J, Johnson PF. Regulation of senescence and the SASP by the transcription factor c/ebp β . *Exp Gerontol*. 2019;128:110752.
- Toutfaire M, Bauwens E, Debacq-Chainiaux F. The impact of cellular senescence in skin ageing: A notion of mosaic and therapeutic strategies. *Biochem Pharmacol*. 2017;142:1–12.
- Franco AC, Martini H, Victorelli S, Lagnado AB, Wyles SP, Rowsey JL, et al. Senescent cell transplantation into the skin induces age-related peripheral dysfunction and cognitive decline. *Aging Cell*. 2025;24(1):e14340.
- Myrianthopoulos V. The emerging field of senotherapeutic drugs. *Future Med Chem*. 2018;10(20):2369–72.
- Pignolo RJ, Passos JF, Khosla S, Tchkonja T, Kirkland JL. Reducing senescent cell burden in aging and disease. *Trends Mol Med*. 2020;26(7):630–8.
- Wang R, Yu Z, Sunchu B, Shoaf J, Dang I, Zhao S, et al. Rapamycin inhibits the secretory phenotype of senescent cells by a Nrf2-independent mechanism. *Aging Cell*. 2017;16(3):564–74.
- Moiseeva O, Deschenes-Simard X, St-Germain E, Igelmann S, Huot G, Cadar AE, et al. Metformin inhibits the senescence-associated secretory phenotype by interfering with IKK/NF- κ B activation. *Aging Cell*. 2013;12(3):489–98.
- Chung CL, Lawrence I, Hoffman M, Elgindi D, Nadhan K, Potnis M, et al. Topical Rapamycin reduces markers of senescence and aging in human skin: an exploratory, prospective, randomized trial. *Geroscience*. 2019;41(6):861–9.
- Banavase Channakeshavaiah R, Andanooru Chandrappa NK. Topical Metformin in the treatment of melasma: A preliminary clinical trial. *J Cosmet Dermatol*. 2020;19(5):1161–4.
- Takaya K, Asou T, Kishi K. New Senolysis approach via Antibody-Drug conjugate targeting of the senescent cell marker Apolipoprotein D for skin rejuvenation. *Int J Mol Sci*. 2023;24(6).
- Hernandez-Segura A, Nehme J, Demaria M. Hallmarks of cellular senescence. *Trends Cell Biol*. 2018;28(6):436–53.
- Wang AS, Dreesen O. Biomarkers of cellular senescence and skin aging. *Front Genet*. 2018;9:247.
- Coppe JP, Desprez PY, Krtolica A, Campisi J. The senescence-associated secretory phenotype: the dark side of tumor suppression. *Annu Rev Pathol*. 2010;5:99–118.
- Sharpless NE, Sherr CJ. Forging a signature of in vivo senescence. *Nat Rev Cancer*. 2015;15(7):397–408.
- Di Micco R, Krizhanovsky V, Baker D, D'Adda Di Fagagna F. Cellular senescence in ageing: from mechanisms to therapeutic opportunities. *Nat Rev Mol Cell Biol*. 2021;22(2):75–95.
- Ren R, Ocampo A, Liu GH, Izpisua Belmonte JC. Regulation of stem cell aging by metabolism and epigenetics. *Cell Metab*. 2017;26(3):460–74.
- Birch J, Barnes PJ, Passos JF. Mitochondria, telomeres and cell senescence: implications for lung ageing and disease. *Pharmacol Ther*. 2018;183:34–49.
- Sun Y, Ma C, Sun H, Wang H, Peng W, Zhou Z, et al. Metabolism: A novel shared link between diabetes mellitus and Alzheimer's disease. *J Diabetes Res*. 2020;2020:4981814.
- Yu EPK, Reinhold J, Yu H, Starks L, Uryga AK, Foote K, et al. Mitochondrial respiration is reduced in atherosclerosis, promoting necrotic core formation and reducing relative fibrous cap thickness. *Arterioscler Thromb Vasc Biol*. 2017;37(12):2322–32.
- Wiley CD, Velarde MC, Lecot P, Liu S, Sarnoski EA, Freund A, et al. Mitochondrial dysfunction induces senescence with a distinct secretory phenotype. *Cell Metab*. 2016;23(2):303–14.
- Mu WC, Ohkubo R, Widjaja A, Chen D. The mitochondrial metabolic checkpoint in stem cell aging and rejuvenation. *Mech Ageing Dev*. 2020;188:11254.
- Wiley CD, Campisi J. The metabolic roots of senescence: mechanisms and opportunities for intervention. *Nat Metab*. 2021;3(10):1290–301.
- Pi C, Yang Y, Sun Y, Wang H, Sun H, Ma M, et al. Nicotinamide phosphoribosyltransferase postpones rat bone marrow mesenchymal stem cell senescence by mediating NAD(+)–Sirt1 signaling. *Aging*. 2019;11(11):3505–22.
- Pi C, Ma C, Wang H, Sun H, Yu X, Gao X, et al. Mir-34a suppression targets Namp1 to ameliorate bone marrow mesenchymal stem cell senescence by regulating NAD(+)–Sirt1 pathway. *Stem Cell Res Ther*. 2021;12(1):271.
- Garcia-Prat L, Sousa-Victor P, Munoz-Canoves P. Proteostatic and metabolic control of stemness. *Cell Stem Cell*. 2017;20(5):593–608.
- Correia-Melo C, Marques FD, Anderson R, Hewitt G, Hewitt R, Cole J, et al. Mitochondria are required for pro-ageing features of the senescent phenotype. *EMBO J*. 2016;35(7):724–42.
- Wilson N, Kataura T, Korsgen ME, Sun C, Sarkar S, Korolchuk VI. The autophagy–NAD axis in longevity and disease. *Trends Cell Biol*. 2023;33(9):788–802.
- Aman Y, Frank J, Lautrup SH, Matysek A, Niu Z, Yang G, et al. The NAD(+)–mitophagy axis in healthy longevity and in artificial intelligence-based clinical applications. *Mech Ageing Dev*. 2020;185:111194.

40. Liu X, Zheng P, Wang X, Dai G, Cheng H, Zhang Z, et al. A preliminary evaluation of efficacy and safety of wharton's jelly mesenchymal stem cell transplantation in patients with type 2 diabetes mellitus. *Stem Cell Res Ther*. 2014;5(2):57.
41. Liao LL, Looi QH, Chia WC, Subramaniam T, Ng MH, Law JX. Treatment of spinal cord injury with mesenchymal stem cells. *Cell Biosci*. 2020;10:112.
42. Li J, Wang T, Hou X, Li Y, Zhang J, Bai W, et al. Extracellular vesicles: opening up a new perspective for the diagnosis and treatment of mitochondrial dysfunction. *J Nanobiotechnol*. 2024;22(1):487.
43. Wu P, Zhang B, Han X, Sun Y, Sun Z, Li L, et al. HucMSC exosome-delivered 14-3-3zeta alleviates ultraviolet radiation-induced photodamage via SIRT1 pathway modulation. *Aging*. 2021;13(8):11542–63.
44. Sun Z, Wang T, Hou X, Bai W, Li J, Li Y, et al. Mesenchymal stromal cells-derived small extracellular vesicles protect against UV-induced Photoaging via regulating pregnancy zone protein. *Stem Cells Transl Med*. 2024;13(11):1129–43.
45. Gong B, Pan Y, Vempati P, Zhao W, Knable L, Ho L, et al. Nicotinamide riboside restores cognition through an upregulation of proliferator-activated receptor-gamma coactivator 1alpha regulated beta-secretase 1 degradation and mitochondrial gene expression in alzheimer's mouse models. *Neurobiol Aging*. 2013;34(6):1581–8.
46. Hong W, Mo F, Zhang Z, Huang M, Wei X. Nicotinamide mononucleotide: A promising molecule for therapy of diverse diseases by targeting NAD + Metabolism. *Front Cell Dev Biol*. 2020;8:246.
47. Qiao C, Xu W, Zhu W, Hu J, Qian H, Yin Q, et al. Human mesenchymal stem cells isolated from the umbilical cord. *Cell Biol Int*. 2008;32(1):8–15.
48. Wang M, Yang D, Li L, Wu P, Sun Y, Zhang X, et al. A dual role of mesenchymal stem cell derived small extracellular vesicles on TRPC6 protein and mitochondria to promote diabetic wound healing. *ACS Nano*. 2024;18(6):4871–85.
49. Ge J, Ye L, Cheng M, Xu W, Chen Z, Guan F. Preparation of microgels loaded with lycopene/nmn and their protective mechanism against acute liver injury. *Food Funct*. 2024;15(2):809–22.
50. Liang W, Chen Q, Cheng S, Wei R, Li Y, Yao C, et al. Skin chronological aging drives age-related bone loss via secretion of cystatin-A. *Nat Aging*. 2022;2(10):906–22.
51. Song Q, Zhou X, Xu K, Liu S, Zhu X, Yang J. The safety and antiaging effects of nicotinamide mononucleotide in human clinical trials: an update. *Adv Nutr*. 2023;14(6):1416–35.
52. Makrantonaki E, Zouboulis CC, William J. Cunliffe scientific awards. Characteristics and pathomechanisms of endogenously aged skin. *Dermatology*. 2007;214(4):352–60.
53. Awad F, Assrawi E, Louvrier C, Jumeau C, Giurgea I, Amselem S, et al. Photoaging and skin cancer: is the inflammasome the missing link? *Mech Ageing Dev*. 2018;172:131–7.
54. Twarda-Clapa A, Olczak A, Bialkowska AM, Koziolkiewicz M. Advanced glycation End-Products (AGEs): formation, chemistry, classification, receptors, and diseases related to ages. *Cells*. 2022;11(8).
55. Chen B, Sun Y, Zhang J, Zhu Q, Yang Y, Niu X, et al. Human embryonic stem cell-derived exosomes promote pressure ulcer healing in aged mice by rejuvenating senescent endothelial cells. *Stem Cell Res Ther*. 2019;10(1):142.
56. Saleh DO, Mansour DF, Hashad IM, Bakeer RM. Effects of Sulforaphane on D-galactose-induced liver aging in rats: role of keap-1/nrf-2 pathway. *Eur J Pharmacol*. 2019;855:40–9.
57. Jafarzadeh A, PourMohammad A, Goodarzi A. A systematic review of the efficacy, safety and satisfaction of regenerative medicine treatments, including platelet-rich plasma, stromal vascular fraction and stem cell-conditioned medium for hypertrophic scars and keloids. *Int Wound J*. 2024;21(4):e14557.
58. Zare S, Jafarzadeh A, Zare S, Shamloo A. Exploring the dermatological applications of human mesenchymal stem cell secretome: a comprehensive review. *Stem Cell Res Ther*. 2025;16(1):177.
59. Jafarzadeh A, Pour Mohammad A, Keramati H, Zeinali R, Khosravi M, Goodarzi A. Regenerative medicine in the treatment of specific dermatologic disorders: a systematic review of randomized controlled clinical trials. *Stem Cell Res Ther*. 2024;15(1):176.
60. Zhang H, Xiao X, Wang L, Shi X, Fu N, Wang S, et al. Human adipose and umbilical cord mesenchymal stem cell-derived extracellular vesicles mitigate Photoaging via TIMP1/Notch1. *Signal Transduct Target Ther*. 2024;9(1):294.
61. Yu CQ, Xu XG, Chen HD, Li YH. Clinical efficacy and safety of nano-microneedle-assisted phenylethyl resorcinol for the treatment of infraorbital dark circles. *J Cosmet Dermatol*. 2021;20(3):884–9.
62. Cao Y, Yang JP, Zhu XG, Zhu J, Chang HQ, Guo SH, et al. A comparative in vivo study on three treatment approaches to applying topical botulinum toxin A for crow's feet. *Biomed Res Int*. 2018;2018:6235742.
63. Suryadevara V, Hudgins AD, Rajesh A, Pappalardo A, Karpova A, Dey AK, et al. SenNet recommendations for detecting senescent cells in different tissues. *Nat Rev Mol Cell Biol*. 2024;25(12):1001–23.
64. Thau H, Gerjol BP, Hahn K, von Gudenberg RW, Knoedler L, Stallcup K, et al. Senescence as a molecular target in skin aging and disease. *Ageing Res Rev*. 2025;105:102686.
65. Zhuang Y, Jiang S, Deng X, Lao A, Hua X, Xie Y, et al. Energy metabolism as therapeutic target for aged wound repair by engineered extracellular vesicle. *Sci Adv*. 2024;10(15):eadl0372.
66. Belenky P, Bogan KL, Brenner C. NAD + metabolism in health and disease. *Trends Biochem Sci*. 2007;32(1):12–9.
67. Qiu X, Brown K, Hirschey MD, Verdin E, Chen D. Calorie restriction reduces oxidative stress by SIRT3-mediated SOD2 activation. *Cell Metab*. 2010;12(6):662–7.
68. Dai SH, Chen T, Wang YH, Zhu J, Luo P, Rao W, et al. Sirt3 protects cortical neurons against oxidative stress via regulating mitochondrial Ca²⁺ and mitochondrial biogenesis. *Int J Mol Sci*. 2014;15(8):14591–609.
69. Wozny MR, Di Luca A, Morado DR, Picco A, Khaddaj R, Campomanes P, et al. In situ architecture of the ER-mitochondria encounter structure. *Nature*. 2023;618(7963):188–92.
70. Janikiewicz J, Szymanski J, Malinska D, Patalas-Krawczyk P, Michalska B, Duszynski J, et al. Mitochondria-associated membranes in aging and senescence: structure, function, and dynamics. *Cell Death Dis*. 2018;9(3):332.

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